Exp 2 Matlab Code

```
clc;
clear;
close all;
% Parameters
Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20*fc;
t = 0:1/fs:4/fm; \
% Signals
m = Am*sin(2*pi*fm*t);            % Message
c = Ac*sin(2*pi*fc*t);            % Carrier
mu = Am/Ac;                       % Modulation index
am = Ac*(1+mu*m).*c;              % AM signal
% Demodulation using Hilbert transformc b
dem = abs(hilbert(am));
dem = dem - mean(dem);            % Remove DC offset
% Frequency domain
N = length(am);
f = linspace(-fs/2,fs/2,N);
AM_FFT = abs(fftshift(fft(am)))/N;
% Sidebands and power
USB = fc + fm;
LSB = fc - fm;
Pc = Ac^2/2;
Psb = (mu^2*Pc)/4;
```

% Plotting

figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1), plot(t,m), title('Message Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on

subplot(3,2,2), plot(t,c), title('Carrier Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on

subplot(3,2,3), plot(t,am), title('AM Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on

subplot(3,2,4), plot(t,dem), title('Demodulated Signal'), xlabel('Time (s)'), ylabel('Amplitude'), grid on

subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Spectrum (Sidebands)'), xlabel('Frequency (Hz)'), ylabel('Magnitude'), grid on

subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}), title('Power Distribution'), ylabel('Power (W)'), grid on

% Print results

fprintf('μ = %.2f | USB = %d Hz | LSB = %d Hz | Pc = %.2f W | Psb = %.3f W\n', mu, USB, LSB, Pc, Psb);

